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DEPARTMENT OF ROBOTICS AND ARTIFICIAL INTELLIGENCE

MINI PROJECT
(Research Experience Through Practice II - 25MRA203)
On

Holonomic Pipe Traversal Robot using Mecanum & Omni Wheels

Submitted in partial fulfilment of M.Tech Degree in
“ROBOTICS AND ARTIFICIAL INTELLIGENCE”

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**DEPARTMENT OF ROBOTICS AND ARTIFICIAL INTELLIGENCE
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DECLARATION

I, the undersigned student **Ashwin Anil (NB25MRA001)** hereby declare that I am fully aware of the **project conduction procedures and evaluation rubrics** as prescribed by the department. I understand that **the project work will be evaluated through continuous assessment** across the defined phases.

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CONFIRMATION BY GUIDE

I, **Mr. Sunil Kumar S**, hereby declare that I shall guide the above-mentioned student in the project entitled “**Holonomic Pipe Traversal Robot using Mecanum & Omni Wheels**” and will ensure that the Mini project work is carried out under my supervision and submitted as per the academic requirements for the academic year **2025–2026**.

Signature of the Guide

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I extend my heartfelt thanks to **Mr. Sunil Kumar H S**, Assistant Professor, Department of Robotics and Artificial Intelligence, NMIT, for his continuous guidance, valuable suggestions, and unwavering encouragement throughout the course of this work. His technical insights and constructive feedback have been instrumental in shaping the direction and quality of this project.

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ABSTRACT

Inspection and maintenance of vertical cylindrical structures such as utility poles, industrial pipelines, and towers are conventionally performed by human workers, posing significant risks of injury, high operational costs, and inefficiencies in reaching structurally complex or elevated regions. This project presents the design and development of a Holonomic Pipe Traversal Robot using Mecanum and Omni Wheels — a robotic platform capable of climbing, traversing, and performing full circumferential movement around cylindrical structures without mechanical repositioning.

The robot employs a ring-type semi-circular clamping frame that surrounds the target pole or pipe, with four Mecanum wheels and eight Omni wheels mounted symmetrically around the frame to maintain continuous and balanced contact with the cylindrical surface. The Mecanum wheels provide driven traction enabling multi-directional (holonomic) motion — vertical climbing, horizontal traversal, and rotational circumnavigation — while the Omni wheels passively ensure lateral stability and balanced force distribution. The mecanum wheels are mounted on a spring-loaded arm mechanism allows the robot to self-adapt to varying pipe diameters, enhancing versatility.

The system uses Dynamixel XC430-W240-T smart servo motors selected based on torque requirement calculations, controlled via an OpenCM-9.04 microcontroller and OpenCM 485 Expansion Board using TTL communication. The structural components were designed in Autodesk Fusion 360 and fabricated through 3D printing for prototyping.

The robot offers a scalable solution for inspection, maintenance, and surveillance applications across utility, industrial, and infrastructure sectors.

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1. INTRODUCTION

1.1 Project Overview

The increasing complexity and scale of modern infrastructure — spanning industrial pipelines, utility transmission towers, communication poles, and cylindrical storage tanks — has created a pressing demand for reliable and efficient inspection and maintenance technologies. These structures are critical to sectors including oil and gas, power transmission, telecommunications, water supply, and civil infrastructure. Ensuring their structural integrity, detecting surface defects, and performing routine maintenance are essential activities to prevent failures, ensure safety, and comply with regulatory standards.

Traditionally, inspection and maintenance of vertical cylindrical structures have relied on human workers using scaffolding, rope access, or aerial lift platforms. These methods are not only time-intensive and expensive but expose workers to considerable risks including falls from height, confined-space hazards, and exposure to hazardous materials. In critical industrial environments such as chemical plants or high-voltage transmission towers, manual intervention carries even greater safety implications.

Robotic systems offer a transformative solution to these challenges. Field robotics and industrial inspection robotics have evolved significantly over the past decade, with robots being deployed in environments that are inaccessible, dangerous, or impractical for human entry. Among these, pole-climbing and pipe-traversal robots represent a specialized and technically demanding class, as they must overcome gravitational forces, maintain stable grip on curved surfaces, and maneuver in constrained three-dimensional spaces.

The domain of Robotic Manipulation and Field Robotics converges in this project, combining principles of mechanical design, wheel-surface interaction mechanics, motor control, and embedded systems to engineer a functional climbing robot. The project also aligns with the broader field of Industrial Inspection Robotics, which seeks to replace manual inspection workflows with automated, repeatable, and data-capable robotic systems.



Figure 1 Traditional Pipe Climbing Method

1.2 Problem Statement

Inspection and maintenance of vertical pipes / poles and cylindrical structures — such as utility poles, communication towers, and industrial pipes — are often performed manually. This approach is time-consuming, physically hazardous, and operationally inefficient, particularly for structures located at heights, in confined environments, or in chemically aggressive surroundings.

Existing robotic solutions for pole climbing are predominantly non-holonomic in nature, relying on inchworm-type mechanisms, wheeled grippers with standard wheels, or magnetic adhesion systems. These designs are largely limited to vertical motion along a single axis, requiring mechanical repositioning or reorientation to traverse circumferentially around a cylindrical structure. Inchworm-based robots suffer from low climbing speed and high energy consumption, while many wheeled designs lack the adaptability to handle varying pole diameters or maintain stable grip under dynamic loading.

Furthermore, current research reveals a significant gap in the development of holonomic pole-climbing robots that employ Mecanum or Omni wheels — wheel types well-established in ground mobile robotics for their ability to enable multi-directional movement without changing vehicle orientation. The application of these wheel technologies to cylindrical climbing contexts remains largely unexplored and underdeveloped.

There is therefore a clear and unmet need for a stable, energy-efficient, and holonomic pipe traversal robot that is capable of vertical climbing, horizontal traversal along the pipe axis, and complete 360° circumferential movement around cylindrical structures — all without the need for mechanical repositioning of the robot's body. Such a system would significantly enhance inspection coverage, reduce operational downtime, and eliminate the need for human workers to perform high-risk climbing tasks.

1.3 Significance of the Problem

The significance of developing a holonomic pipe traversal robot extends across multiple dimensions:

- **Safety:** Eliminating the need for human workers to climb poles and pipelines directly reduces the risk of work-related injuries and fatalities. In sectors such as oil refinery inspection and high-voltage transmission line maintenance, this safety benefit is of paramount importance.
- **Operational Efficiency:** A robot capable of continuous vertical and circumferential traversal can complete inspection runs faster than manual crews, with no rest requirements and with the ability to carry sensors or cameras for real-time data acquisition.
- **Coverage and Accuracy:** Holonomic movement allows the robot to reach any point on the cylindrical surface — top, side, or underside — without dismounting and repositioning, enabling complete structural coverage and reducing the probability of missed defects.
- **Scalability and Adaptability:** A system designed with adjustable clamping and modular architecture can be deployed across a range of pipe diameters and structural types, making it applicable to a broad range of industries and infrastructure types.
- **Technological Advancement:** Addressing the identified research gap in holonomic climbing robotics contributes to the broader advancement of the field, offering a new paradigm for pipe and pole traversal that can be further developed for autonomous inspection, payload delivery, or remote sensing applications.

2. OBJECTIVES OF THE PROJECT

The following objectives define the scope and intended outcomes of this project:

1. **To design and develop a holonomic pipe traversal robot** using Mecanum and Omni wheels arranged symmetrically around a ring-type clamping frame, capable of maintaining stable and continuous contact with the outer surface of vertical cylindrical pipes.
2. **To achieve full three-dimensional surface traversal** — including vertical climbing along the pipe axis, horizontal translation across pipe length, and complete 360° circumferential movement around the pipe cross-section — without mechanical repositioning or interruption of motion.
3. **To implement a self-adaptive clamping mechanism** using spring-loaded telescopic arms that automatically adjusts to varying pipe diameters, enabling the robot to be deployed across a range of standard industrial pipe sizes without hardware reconfiguration.
4. **To develop and validate a synchronized motor control system** using Dynamixel XC430 smart servo motors and the OpenCM-9.04 controller, ensuring smooth, stable, and precisely coordinated wheel motion during all traversal modes on the pipe surface.
5. **To establish a scalable and modular robotic platform** that serves as a functional proof-of-concept for external pipeline inspection, providing a foundation for the future integration of NDT sensors, cameras, or wireless telemetry for real-world industrial inspection applications.

3. LITERATURE SURVEY AND REVIEW

Table 1 Literature Survey

Paper Title	Authors	Publication (Journal / Conference)	Year	Methodology	Summary
Research Progress on Pole-Climbing Robots: A Review	D. Tang et al.	Recent Patents on Engineering	2024	Review of wheeled, tracked, magnetic, and hybrid climbing robot mechanisms.	Discusses recent developments, classifications, and challenges in pole-climbing robot stability and adaptability.
Pipe Traversing Apparatus and Methods	J. Ayus et al.	US Patent – US11732832B2	2023	Pipe traversal robot using articulated wheel assemblies and adaptive gripping arms.	Improves stability and traversal across straight, curved, and junction pipe sections.
Modular Pipe Traversing Apparatus and Methods	J. Ayus et al.	US Patent – US12252198B2	2025	Modular pipe-climbing robot with omnidirectional wheel modules and adaptive frame structure.	Enhances maneuverability, scalability, and traversal on pipes of varying diameters.
Advances in Climbing Robots for Vertical Structures: A Review	G. Fang et al.	Biomimetics	2023	Review of bio-inspired climbing and adhesion mechanisms.	Compares locomotion and gripping methods for vertical climbing applications.
Development of a Wheel-Leg Hybrid Climbing Robot with Switchable Permanent Magnetic Omni-Wheels as Feet	Y. Bu et al.	IEEE ROBIO Conference	2022	Hybrid wheel-leg robot using magnetic omni-wheels for climbing.	Improves obstacle negotiation and climbing adaptability on metallic surfaces.
Advances in Hybrid Modular Climbing Robots: Design Principles and Refinement Strategies	R. Poon & I. Hunter	arXiv Preprint	2025	Modular climbing robot architecture with reconfigurable mechanisms.	Focuses on adaptability, scalability, and performance optimization.
Design and Development of a Low-Cost Pole Climbing Robot Using Arduino Mega	H. Khairam et al.	Journal of Physics: Conference Series	2021	Low-cost wheeled climbing robot using Arduino-based control.	Demonstrates a simple and affordable climbing mechanism for inspection purposes.

Review of Advancements in Wall and Vertical Climbing Robot Techniques	J. Zhu et al.	Robotics and Autonomous Systems	2024	Review of wall-climbing and adhesion technologies.	Compares magnetic, vacuum, and gripping mechanisms used in climbing robots.
Design and Optimization of the Magnetic Adsorption Mechanism of a Climbing Robot	L. Xinrui et al.	Journal of Mechanical Science and Technology	2021	Optimization of magnetic adhesion force and wheel arrangement.	Improves load capacity and climbing stability using magnetic adsorption.
A Review on Wheeled Type In-Pipe Inspection Robot	Rajendran Sugin Elankavi et al.	International Journal of Mechanical Engineering and Robotics Research	2022	Review of wheeled in-pipe inspection robot mechanisms.	Discusses adaptive wheel systems, wall pressing, and pipe traversal methods.
Development of Pipe Robot by Using Mecanum Wheels	Daeyoung Kim et al.	Korean Society of Manufacturing Process Engineers	2021	Pipe robot using mecanum wheels for omnidirectional motion.	Improves maneuverability and traversal through bends and vertical pipes.
Development of a Wheel-Type In-Pipe Robot Using Slider-Crank and CVT Mechanisms	J. Park et al.	Biomimetics (MDPI)	2024	Adaptive wheel-type robot using slider-crank and CVT mechanisms.	Enables stable movement through varying pipe diameters and curved sections.
Design and Analysis of Modular Pipe Climber-III with a Multi-Output Differential Mechanism	Vishnu Kumar et al.	Robotics Research / arXiv	2021	Modular pipe climber with passive differential torque distribution.	Reduces wheel slippage and improves bend traversal stability.
Locomotion Methods of Pipe Climbing Robots: A Review	P. Chattopadhyay et al.	Journal of Engineering Science and Technology Review	2018	Review of wheeled, tracked, snake-like, and hybrid locomotion systems.	Provides comparison of pipe climbing locomotion techniques and limitations.
Design and Manufacturing of a Mecanum Wheel for Pipe Climbing Robot	S.D. Kamdar et al.	Research Paper	2015	Design optimization and fabrication of mecanum wheels for climbing robots.	Improves wheel traction, directional control, and climbing efficiency.
Design and Modelling of Pipe Climbing Robot (PCR)	Rakesh Rajendran et al.	SSRG International Journal of Mechanical Engineering	2026	Simulation and modeling of adaptive pipe-climbing robot mechanisms.	Discusses gripping force analysis and traversal performance improvements.

4. LIMITATIONS OF THE EXISTING WORK (GAP ANALYSIS)

Existing pipe-climbing and pipe-traversal robots have demonstrated effective locomotion for inspection and maintenance applications; however, several limitations still exist in current designs and research works.

- Many existing robots are designed only for either vertical climbing or circumferential movement, but not both simultaneously with stable holonomic motion.
- Several systems rely on complex magnetic or vacuum adhesion mechanisms, limiting their applicability to specific pipe materials and increasing system cost and power consumption.
- Existing climbing robots often use rigid wheel mounting structures, resulting in poor adaptability to surface irregularities and variations in pipe diameter.
- Some mechanisms experience wheel slippage and unstable gripping due to uneven force distribution and insufficient adaptive tension control.
- Modular and hybrid climbing robots proposed in literature are often mechanically complex, bulky, and difficult to manufacture for low-cost prototype development.
- Limited research is available on compact pole-climbing robots combining Mecanum wheels, Omni wheels, spring-loaded hinged mechanisms, and adaptive gripping in a single integrated design.

Many current systems focus primarily on industrial pipelines and lack simplified designs suitable for educational, research, and low-cost inspection applications.

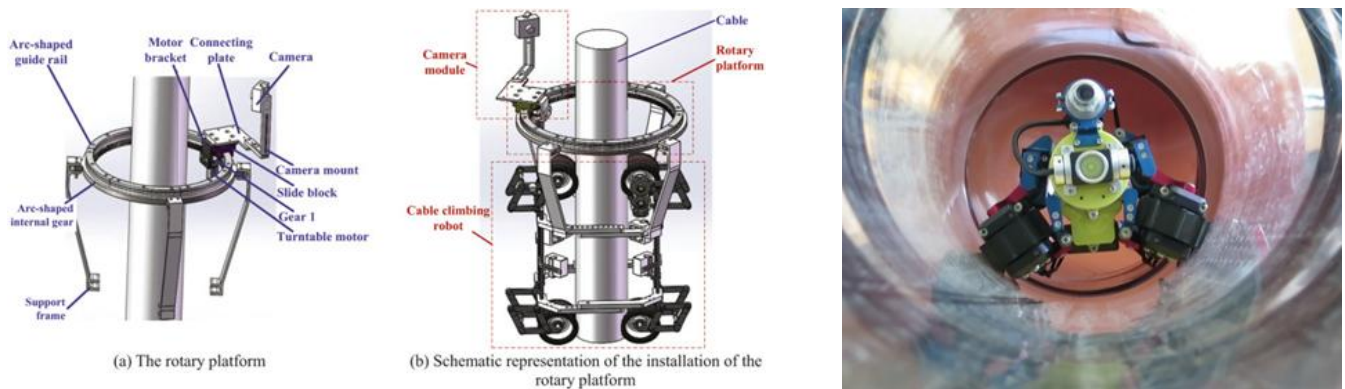


Figure 2 Existing Pipe Traversal Solutions

5. FUNCTIONAL REQUIREMENTS FOR THE PROTOTYPE

The following functional requirements define the core capabilities that the prototype system is expected to demonstrate during the Mini Project Phase 1 evaluation. These requirements are scoped to the primary success use cases and are framed as SMART (Specific, Measurable, Achievable, Relevant, Time-bound) statements.

The prototype for this phase is a 3D-printed ring-frame pipe-gripping robot equipped with Mecanum and Omni wheels, driven by Dynamixel XC430 smart servo motors and controlled via the OpenCM-9.04 microcontroller. The emphasis is on demonstrating stable pipe gripping, controlled vertical climbing, and initiation of circumferential traversal on a fixed-diameter test pipe under laboratory conditions.

Table 2 Functional Requirements Table

Req ID	Functional Requirement (SMART)	SMART Justification	Expected Outcome
FR1	The robot shall grip and hold its position on a vertical cylindrical pipe without slipping under its own weight.	Specific: Stable grip on vertical pipe Measurable: second hold without slippage Achievable: Core mechanical function of prototype Relevant: Foundational requirement for all traversal operations Time-Bound: Demonstrated in Mini Project evaluation	The robot remains securely clamped on the test pipe at any mounted height without downward drift, confirming sufficient clamping force and wheel-to-surface contact.
FR2	The robot shall climb vertically upward and downward along a cylindrical pipe	Specific: Vertical climbing in both directions Measurable: distance travelled by the robot on the pipe Achievable: Primary locomotion mode of the prototype Relevant: Directly addresses pipe traversal along the axial direction Time-Bound: Demonstrated in Phase 1 evaluation	The robot ascends and descends the test pipe in a controlled, straight trajectory without slippage or misalignment, validating vertical locomotion capability.

FR3	The robot shall rotate circumferentially around the outer surface of a vertical cylindrical pipe in both clockwise and counterclockwise directions without vertical drift.	Specific: Circumferential rotation in both directions Measurable: Minimum 90° rotation with vertical drift within ± 10 mm Achievable: Enabled by holonomic wheel arrangement in prototype Relevant: Core capability differentiating holonomic from non-holonomic climbing Time-Bound: Demonstrated in Phase 1 evaluation	The robot rotates around the pipe cross-section in both directions while maintaining its axial position, confirming holonomic movement capability on a cylindrical surface.
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6. PROPOSED METHODOLOGY FOR BUILDING PROTOTYPE

This section outlines the planned approach for designing, developing, and validating the Holonomic Pipe Traversal Robot prototype within the Mini Project scope. The methodology is structured as a sequential workflow, progressing from conceptual design through fabrication to functional validation, ensuring that each stage builds upon the previous one in a logical and manageable manner.

6.1 Phase 1 — Requirement Analysis and Design Specification

The first phase involves consolidating the functional requirements identified in Section Chapter 5 into concrete design specifications. This includes determining the target pipe diameter range, estimating the robot's total weight and payload, and defining the minimum performance benchmarks for each traversal mode. Based on these specifications, the key design parameters — clamping force, wheel contact geometry, motor torque requirement, and frame dimensions — will be analytically derived to guide all subsequent design decisions.

A free-body diagram analysis of the forces acting on a Mecanum wheel pressed against a vertical cylindrical pipe surface will be carried out to calculate the minimum torque required per motor, accounting for the robot's weight, normal contact force, and the coefficient of friction between the wheel and pipe material.

6.2 Phase 2 — Mechanical Design and CAD Modelling

Following design specification, the complete 3D CAD model of the robot will be developed in **Autodesk Fusion 360**. This phase covers the detailed design of:

- The ring-type semi-circular clamping frame that surrounds the pipe
- Wheel mounting brackets for both Mecanum and Omni wheels
- The spring-loaded telescopic arm mechanism for passive diameter adaptation
- Motor housing and shaft coupling arrangements

Multiple design concepts will be evaluated for structural integrity, symmetry of force distribution, and ease of assembly. The finalised concept will be taken forward for fabrication. Design iterations will be carried out within the CAD environment before committing to physical fabrication, minimising material wastage and rework.

6.3 Phase 3 — Fabrication and Assembly

The structural components of the robot frame, wheel mounts, and telescopic arm assemblies will be fabricated using **FDM 3D printing** with PLA/PETG material, as this allows rapid prototyping, design iteration, and cost-effective production within the project timeline. Purchased components — including Mecanum wheels, Omni wheels, Dynamixel motors, fasteners, and electronic modules — will be procured in parallel with the fabrication process to avoid delays.

Assembly will proceed in the following sequence:

1. Frame halves assembled and joined at the central hinge

2. Telescopic spring arms fitted to the frame
3. Mecanum and Omni wheels mounted on their respective brackets and attached to the frame
4. Dynamixel motors coupled to the Mecanum wheel shafts
5. Electronics bay mounted on the frame for controller and power module housing

6.4 Phase 4 — Electronics Integration and Control Setup

Once the mechanical assembly is complete, the electronics will be integrated. The OpenCM-9.04 microcontroller will be mounted on the OpenCM 485 Expansion Board and connected to all four Dynamixel XC430-W240-T motors via the TTL communication bus. The power supply chain — from the 4S LiFePO4 battery through the LM2596 buck converters to the controller and motors — will be wired and verified.

Motor parameters including ID assignment, baud rate, operating mode, and velocity limits will be configured using **Dynamixel Wizard 2.0** via the U2D2 USB interface prior to integration into the full control loop. Each motor will be bench-tested individually to confirm correct communication and response before being integrated into the assembled robot.

6.5 Phase 5 — Control Software Development and Motion Programming

With the hardware fully assembled and electronics verified, motion control software will be developed on the **OpenCM 9.0** in Embedded C/C++. The control program will implement velocity commands for each of the three primary traversal modes defined in the functional requirements — vertical climbing, and circumferential rotation — by writing goal velocity values to each motor's control table register via Dynamixel Protocol 2.0.

Motion routines for each traversal mode will be developed and tested independently before being integrated into a unified control program. The closed-loop velocity feedback available from the XC430-W240-T's built-in encoders will be utilised to monitor and correct motor performance in real time during traversal.

6.6 Phase 6 — Testing and Validation

The final phase involves systematic testing of the assembled and programmed prototype against the functional requirements defined in Chapter 5. Testing will be carried out on a fixed-diameter vertical test pipe under laboratory conditions, following the sequence below:

1. **Static Grip Test (FR1):** Verify that the robot holds its position on the pipe under its own weight without slippage.
2. **Vertical Climbing Test (FR2):** Verify controlled upward and downward climbing.
3. **Circumferential Rotation Test (FR3):** Verify clockwise and counterclockwise rotation of at least 90° with axial drift within ± 10 mm.

7. IMPLEMENTATION DETAILS

This chapter presents the subsystem-wise implementation details of the Holonomic Pipe Traversal Robot prototype, covering the mechanical design evolution, electronic architecture, software control logic, and final integrated system.

7.1 Mechanical Subsystem

The mechanical subsystem forms the physical foundation of the robot — encompassing the clamping frame, wheel arrangement, motor mounting, and the adaptive gripping mechanism. The design evolved through multiple conceptual iterations before arriving at the fabricated prototype.

7.1.1 Conceptual Design – 1

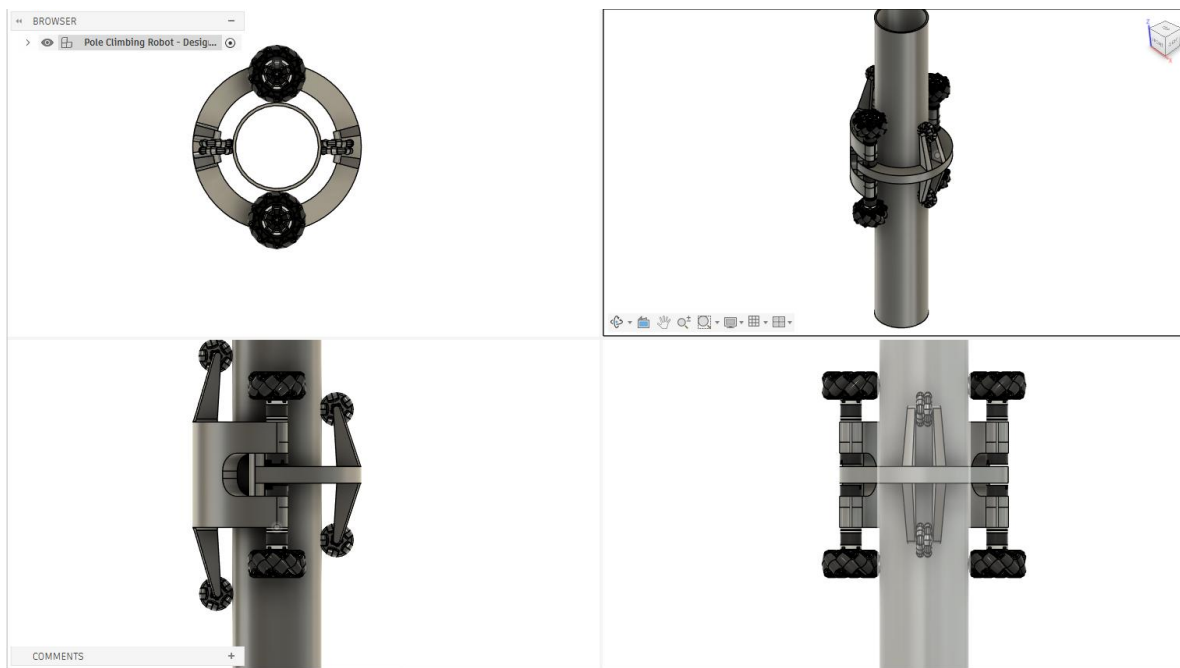


Figure 3 CAD model of Conceptual Design-1

The first concept established the fundamental ring-type architecture for the pipe traversal robot. The CAD model shown in Figure 3, features a circular frame that fully surrounds the cylindrical pipe, consisting of two semi-circular clamp sections connected by a central support structure. Four Mecanum wheels and four Omni wheels are mounted symmetrically around the inner circumference of the frame, maintaining continuous and evenly distributed contact with the pipe surface. The Mecanum wheels serve as the driven elements, generating traction for vertical climbing and circumferential traversal, while the Omni wheels act as passive stabilisers — together with the circular chassis geometry, they ensure balanced radial force distribution around the pipe cross-section. The symmetric wheel arrangement improves stability during both vertical and horizontal traversal, and the overall design enables the robot to grip the pipe firmly while allowing controlled motion along and around the cylindrical surface.

Limitations in this design:

- **Impractical latching mechanism** — the two semi-circular sections were connected by a simple central support structure that did not provide a reliable, repeatable, or secure latching interface, making consistent clamping force difficult to achieve during mounting and operation.
- **Fixed pipe diameter** — the rigid circular frame is sized for a specific pipe diameter with no mechanism to accommodate variation, restricting the robot to a single pipe size.
- **Inconsistent normal force** — with wheels mounted at fixed radial positions on a rigid frame, any dimensional tolerance in the pipe or frame geometry results in uneven contact pressure across wheel positions, leading to differential traction and potential slippage.
- **No diameter adaptability** — the absence of any spring-loaded or adjustable clamping mechanism means the robot cannot self-compensate for pipe surface irregularities or ovality, reducing reliability on real-world pipeline surfaces.

7.1.2 Conceptual Design – 2

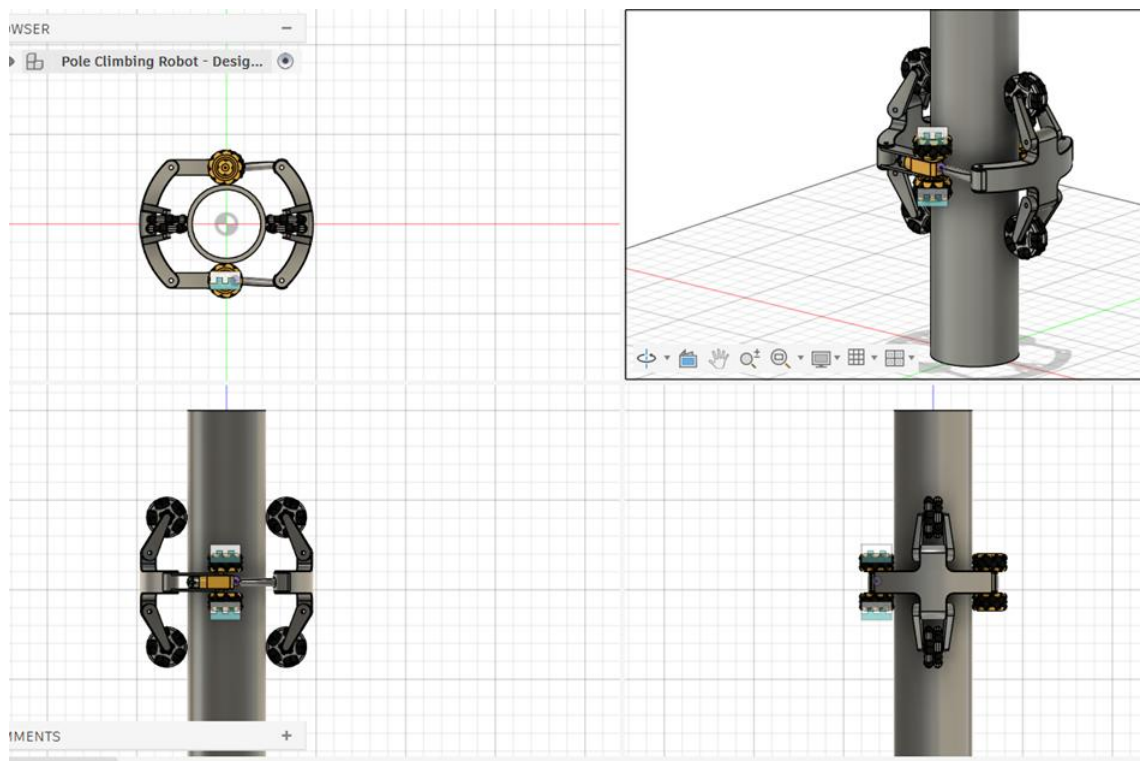


Figure 4 CAD Model of Conceptual Design-2

The second concept retained the core ring-type architecture and symmetric wheel arrangement established in Conceptual Design 1, while introducing targeted mechanical refinements to address the limitations identified in the previous iteration.

The most significant structural update was the introduction of hinged wheel arm assemblies. Rather than mounting the wheel brackets rigidly to the frame, each wheel assembly is attached to a hinged arm that allows slight angular adjustment during operation.

Together, the hinged arm geometry and spring-loaded joints form an **adaptive gripping mechanism** that enables the robot to firmly embrace the pipe while maintaining smooth and stable motion — a significant functional improvement over the fixed-position wheel arrangement of Conceptual Design 1.

Despite these improvements, the concept presented the following limitations:

- **Design refinement required for practical implementation** — while the spring-loaded hinged arm concept was sound in principle, translating it into a physically realizable assembly demanded detailed design work including precise spring rate selection, arm pivot geometry, and wheel contact angle optimization to ensure consistent normal force across all wheel positions.
- **Latch and hinge robustness** — the hinge and locking latch interface lacked sufficient structural rigidity to reliably withstand dynamic loads during active traversal.

7.1.5 Prototype – 1



Figure 5 Prototype 1 mounted on a 4-inch pipe, showing the semi-circular clamping frame, Mecanum and Omni wheel arrangement, and onboard electronics

The first physical prototype was designed in Autodesk Fusion 360 and fabricated using FDM 3D printing with PLA material. The prototype was designed for a **4-inch pipe diameter**, serving as the initial testbed to validate the ring-type clamping concept and holonomic wheel arrangement in a physical assembly.

The wheel configuration comprised of four Mecanum wheels as the primary driven elements, responsible for generating traction for vertical climbing and circumferential traversal, and two Omni wheels — one positioned on the upper half and one on the lower half of the clamping frame — acting as passive stabilisers to maintain balanced contact and prevent axial tilting of the frame during motion.

For the control electronics, an **Arduino Nano** was used as the microcontroller and an **L298N motor driver module** was employed to drive the Mecanum wheel motors, providing bidirectional speed control through PWM signals. This electronics stack was selected for its low cost and rapid prototyping convenience, with

the primary goal of validating the mechanical geometry and wheel-to-pipe contact behavior at an early stage before investing in the higher-cost Dynamixel-based electronics architecture.

Observations from Prototype-1 testing:

- **Two Omni wheels insufficient for stability** — with only one Omni wheel on each half of the frame, lateral stability was limited, and the frame exhibited minor rocking about the pipe axis under dynamic loading.
- **Insufficient motor torque** — the TT gear motors lacked the torque required to overcome the robot's weight during vertical climbing, causing the robot to stall on the pipe surface and establishing the need for a higher-torque actuator in subsequent iterations.
- **L298N motor driver limitations** — the L298N driver introduced noticeable power loss due to voltage drop across its internal transistors, reducing available torque at the wheel and limiting climbing performance, particularly under the combined weight of the electronics and frame.
- **Wiring management** — the absence of a dedicated electronics bay resulted in loose wiring that interfered with moving parts and complicated testing.

7.1.6 Prototype – 2



Figure 6 Prototype 2 mounted on a 4-inch pipe, with the updated four Omni wheel configuration

Prototype 2 was designed and 3D printed for a **4-inch pipe diameter**, retaining the semi-circular clamping frame geometry from Prototype 1 while incorporating a refined wheel configuration based on the observations from the previous iteration. The primary mechanical change in this prototype was the upgrade to a **4 Omni wheel arrangement** — with two Omni wheels positioned on the upper half and two on the lower half of the clamping frame, each mounted at an angle to the pipe surface to maximise radial contact and improve the distribution of gripping force around the pipe circumference.

The electronics stack remained unchanged from Prototype 1 — Arduino Nano, TT gear motors, and L298N motor driver — as the focus of this iteration was specifically to validate the revised wheel configuration before addressing the actuation system.

Observations from Prototype-2 testing:

- **Improved stability and surface contact** — the four Omni wheel configuration produced a noticeably more stable and balanced grip on the pipe surface compared to Prototype 1, with the angled mounting geometry ensuring consistent radial contact pressure across all wheel positions throughout the clamping frame.
- **Lateral slip eliminated** — the addition of the two extra Omni wheels effectively prevented lateral slip on the pipe surface, confirming that the revised wheel arrangement provided adequate circumferential constraint during attempted traversal.

The Prototype 2 testing conclusively established that the wheel geometry and clamping architecture were sound, and that the remaining barrier to successful vertical climbing was purely the actuator torque. This directed the focus of the next phase toward a systematic **motor torque requirement calculation** to determine the minimum torque per motor needed for reliable vertical climbing.

7.1.3 Motor Torque Calculation

Before proceeding to detailed design and prototype, the minimum motor torque required per driven wheel was analytically determined using a free-body diagram of the robot (shown in Figure 5) on a vertical pipe surface.

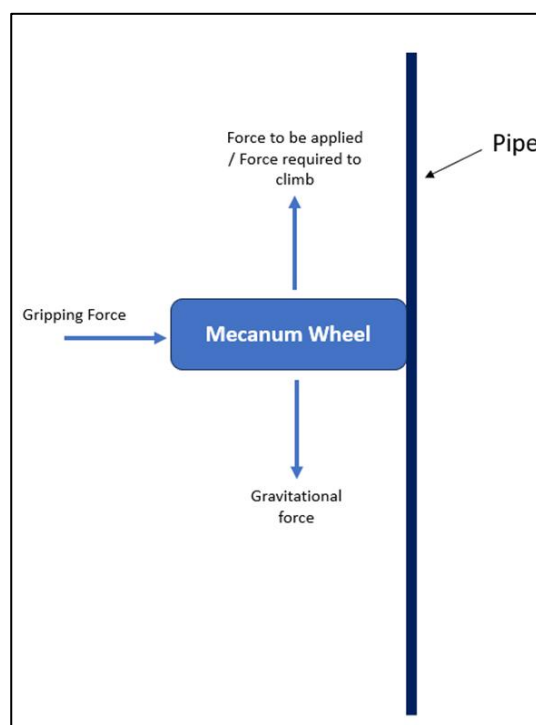


Figure 7 A free-body diagram illustrating the vertical forces acting on a Mecanum wheel as it interacts with a pole

1) Known Parameters

- Payload = 2 kg
- Robot body = 1 kg
- Total mass = 3 kg
- Number of motors = 4
- Wheel diameter = 100 mm
→ radius = 0.05 m

2) Forces Acting on the Robot

For vertical climbing, the robot experiences:

- i. Gravitational / Gradient Resistance Force
- ii. Rolling Resistance Force
- iii. Acceleration Force

a) Gravitational Force / Gradient Resistance

The major opposing force during vertical climbing is gravity.

Gravitational force:

$$F_g = m \cdot g = 3 \times 9.81 = 29.43 \text{ N}$$

b) Rolling Resistance Force

Rolling resistance is:

$$F_{rr} = C_{rr}mg$$

Assume rolling resistance coefficient:

$$C_{rr} = 0.02$$

$$F_{rr} = 0.02 \times 3 \times 9.81$$

$$F_{rr} = 0.5886 \text{ N}$$

c) Acceleration Force

Acceleration force:

$$F_a = ma$$

Assume slow climbing acceleration:

$$a = 0.5 \text{ m/s}^2$$

$$F_a = 3 \times 0.5$$

$$F_a = 1.5 \text{ N}$$

3) Total Tractive Force

The total tractive force required is the sum of all resisting forces:

$$F_{tractive} = F_g + F_{rr} + F_a$$

$$F_{tractive} = 29.43 + 0.5886 + 1.5$$

$$F_{tractive} = 31.52 \text{ N}$$

Force Required to Climb (Including Safety Factor)

Using a safety factor of 1.5:

$$F_{total} = 31.52 \times 1.5$$

$$F_{total} \approx 47.28 \text{ N}$$

4) Tractive Torque

Total torque required:

$$T = Fr$$

$$T_{total} = 47.28 \times 0.04$$

$$T_{total} = 1.8912 \text{ Nm}$$

Torque Per Motor

Since 4 motors are used:

$$T_{motor} = \frac{1.8912}{4}$$

$$T_{motor} = 0.473 \text{ Nm}$$

Motor Safety Margin

Applying an additional motor safety factor of 1.5:

$$T_{required} = 0.473 \times 1.5$$

$$T_{required} \approx \mathbf{0.71 \text{ Nm per motor}}$$

5) Gripping Force Calculation

For vertical climbing, the **normal force** is the gripping force acting perpendicular to the pipe surface.

The **tractive force** is the frictional force generated between the wheel and pipe surface.

The traction relationship is:

$$F_{traction} = \mu N$$

where:

μ = coefficient of friction

N = gripping (normal) force

Assume:

$$\mu = 0.2$$

To avoid slipping during climbing (including safety factor):

$$F_{total} = \mu N \times 1.5$$

Substituting values:

$$47.28 = 0.2 \times N \times 1.5$$

$$N = \frac{47.28}{0.2 \times 1.5}$$

$$N = 157.6 \text{ N}$$

Based on the torque requirement calculation, the Dynamixel XC430-W240-T actuator was selected.

7.1.4 Torsion Spring Selection Calculation

From previous calculation:

$$N = 157.6 \text{ N}$$

This is the total gripping force required.

There are 2 spring-loaded arms, so each arm provides half the gripping force

$$F_{arm} = \frac{157.6}{2}$$

$$F_{arm} = 78.8 \text{ N}$$

1. Spring Torque Requirement

Distance from hinge to wheel center:

$$L = 0.20 \text{ m}$$

Required spring torque:

$$T = FL$$

$$T = 78.8 \times 0.20$$

$$T = 15.76 \text{ Nm}$$

So, each torsion spring should provide approximately:

$$T_{spring} \approx 16 \text{ Nm}$$

2. Angular Deflection Assumption

Assume arm opening during mounting:

$$\theta = 30^\circ$$

$$\theta = 0.524 \text{ rad}$$

3. Required Spring Rate

Spring rate:

$$k = \frac{T}{\theta}$$

$$k = \frac{16}{0.524}$$

$$k \approx 30.53 \text{ Nm/rad}$$

or

$$k \approx 0.5328 \text{ Nm/deg}$$

7.1.7 Final Developed Model

The final developed model represents the culmination of the iterative design and prototyping process, incorporating all mechanical refinements and actuator upgrades identified through Prototype 1, Prototype 2, and the motor torque requirement analysis. The robot was designed and 3D printed for an **8-inch pipe diameter**, with the frame geometry, wheel positions, and clamping mechanism all optimized for this target pipe size.

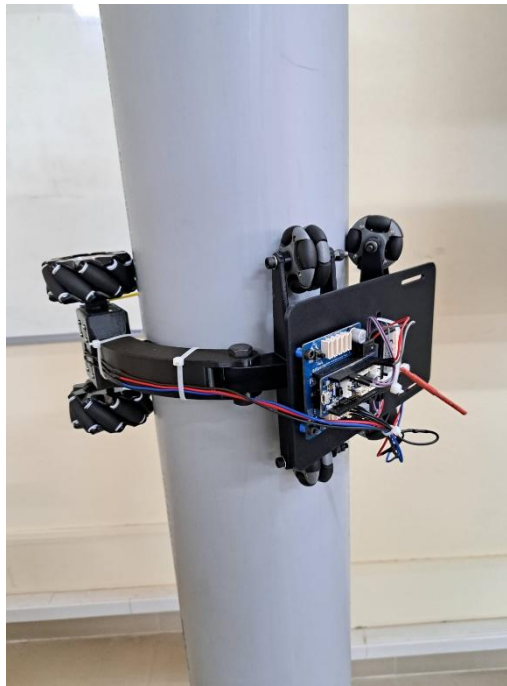


Figure 8 Final developed model mounted on the 8-inch pipe, showing the Dynamixel motor assemblies, Omni wheel arrangement, and integrated electronics bay.

The model was fabricated using FDM 3D printing, incorporating the following key developments:

1) Dynamixel XC430-W240-T Smart Actuator

The TT gear motors used in the earlier prototypes were replaced with **Dynamixel XC430-W240-T** smart servo motors, selected based on the torque requirement calculation which established the minimum torque per motor needed for reliable vertical climbing.



Figure 9 Dynamixel XC430-W240-T smart servo actuator

The XC430-W240-T offers a stall torque of 1.9 Nm at 12V — along with built-in closed-loop velocity and position control via Dynamixel Protocol 2.0, eliminating the need for an external motor driver and substantially improving control precision and reliability over the TT motor and L298N setup used previously.

2) Custom designed and 3D Printed Dynamixel Motor to mecanum wheel Coupling

A **custom Dynamixel motor-to-Mecanum wheel coupling** was designed and 3D printed to interface the XC430-W240-T output shaft with the Mecanum wheel hub. The coupling was designed to match the horn geometry of the Dynamixel motor on one side and the bore of the Mecanum wheel on the other, ensuring concentric and backlash-free power transmission.



Figure 10 3D printed Dynamixel XC430-W240-T motor-to-Mecanum-wheel coupling

Key Features of the Final Model developed:

- Redesigned semi-circular clamping frame for 8-inch pipe and for the mounting of the Dynamixel XC430-W240-T motors.
- Four Mecanum wheels driven by individual Dynamixel motors, each independently addressable over the TTL bus.
- Four Omni wheels retained from Prototype 2 for lateral stability and balanced contact force distribution
- OpenCM-9.04 microcontroller and OpenCM 485 Expansion Board mounted on an integrated electronics plate fixed to the frame
- RC receiver (FSR9B) mounted on the frame for wireless velocity command input
- 12V 6Ah LiFePO4 battery providing power for onboard electronics.

7.2 Electronics Subsystem

The electronic subsystem manages power distribution, motor communication, and control signal acquisition. It consists of four functional blocks: Power, Input, Control, and Actuation.

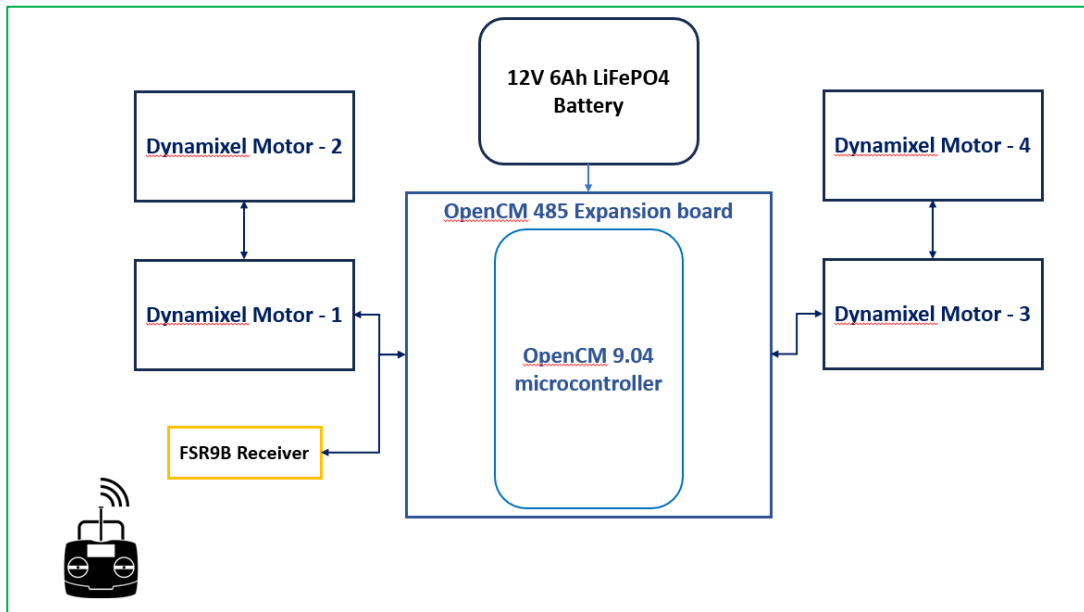


Figure 11 Electronics Block Diagram

The block diagram in Figure 6, illustrates the complete electronic architecture of the pipe traversal robot. The system is organised around the **OpenCM 9.04 microcontroller** mounted on the **OpenCM 485 Expansion Board**, which together form the central control and communication unit of the robot.

- **Power Supply:** The **12V 6Ah LiFePO4 battery** supplies power directly to the OpenCM 485 Expansion Board, which distributes regulated voltage to the microcontroller and provides the motor supply rail for all four Dynamixel motors connected on the RS-485 bus.
- **Input — Wireless Control:** The **FS-R9B RC receiver** is connected to the OpenCM 9.04 and receives PWM signals transmitted wirelessly from the RC handheld transmitter. Two channels are used — Channel 2 for vertical climb commands and Channel 4 for circumferential rotation commands — which are read by the microcontroller using `pulseIn()` on dedicated digital input pins.
- **Control Processing:** The OpenCM 9.04 reads the two incoming PWM signals every 20 ms, converts them to RPM values through the `pwmToRPM()` function, applies the velocity mixing logic, and generates independent left-side and right-side velocity commands.
- **Actuation — Motor Layout:** The four Dynamixel XC430-W240-T motors are connected in two pairs of daisy-chain topology on the TTL bus. **Motors 1 and 2** are assigned to the left side of the clamping frame and receive the `leftVelocity` command; **Motors 3 and 4** are assigned to the right side and receive the `rightVelocity` command. The bidirectional arrows between the controller and the motors indicate that communication is two-way — the controller not only sends goal velocity commands but also reads back present velocity and status data from each motor, enabling closed-loop monitoring.

This architecture results in a compact, fully self-contained electronic system where all sensing, processing, and actuation are integrated within the robot frame with a single wireless control interface.

7.3 Software Subsystem

7.3.1 Overview



Figure 12 OpenCM9.04-C Microcontroller

The control software runs on the OpenCM-9.04 microcontroller and is written in **Embedded C/C++** using the **Arduino IDE**. It uses the **Dynamixel2Arduino library** for communication with the XC430-W240-T motors via Dynamixel Protocol 2.0 over the TTL bus.

*The full source code is provided in **Annexure A**.

7.3.2 Data Flow and Processing Logic

The software follows a linear sense → process → actuate loop executed every 20 ms:

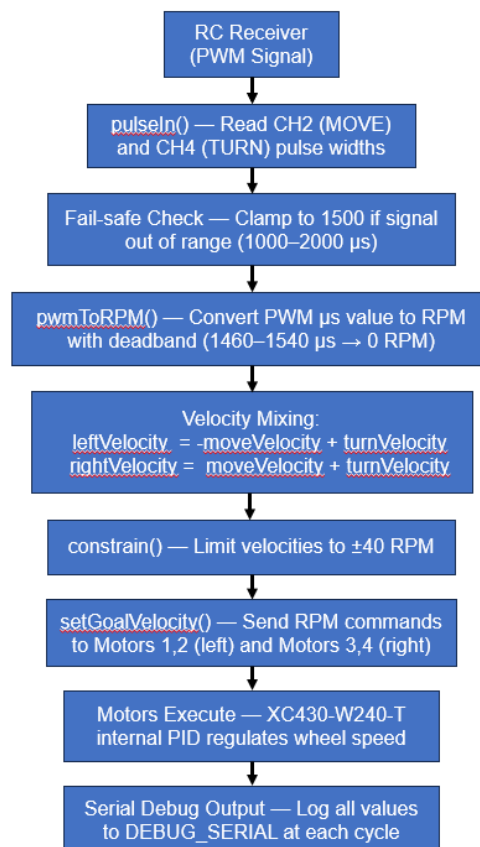


Figure 13 Software Execution Flowchart

7.3.3 Key Functions and Logic Explained

pwmToRPM(int pwm)

This function converts the raw PWM pulse width received from the RC receiver (ranging 1000–2000 μ s, centred at 1500 μ s) into a target RPM value in the range of –40 to +40 RPM. A deadband of ± 40 μ s around the centre (1460–1540 μ s) maps to 0 RPM, preventing motor jitter caused by RC transmitter stick drift or signal noise when the stick is at neutral.

Velocity Mixing Logic

The two RC channels are mixed into left-side and right-side velocity commands using a differential drive analogy adapted for pipe traversal:

Table 3 Velocity mixing logic showing motor command combinations for each traversal mode

	moveVelocity	turnVelocity	leftVelocity	rightVelocity	Robot Motion
Climb Up	+ve	0	–ve	+ve	Vertical ascent
Climb Down	–ve	0	+ve	–ve	Vertical descent
Rotate CW	0	+ve	+ve	+ve	Clockwise rotation
Rotate CCW	0	–ve	–ve	–ve	Counter-clockwise rotation

Fail-safe

If the PWM signal falls outside the valid range of 1000–2000 μ s — which occurs when the RC signal is lost or the receiver is disconnected — the value is forced to 1500 μ s (neutral), resulting in 0 RPM commands to all motors. This prevents uncontrolled robot motion in the event of signal loss.

8. RESULTS AND ANALYSIS

8.1 Overview

The final prototype of the Holonomic Pipe Traversal Robot was assembled and tested against the three functional requirements defined in Chapter 5. Testing was conducted on a 8-inch cylindrical PVC pipe under laboratory conditions. The following section documents the outcomes of each test and the key observations recorded during validation.

8.2 Test Results

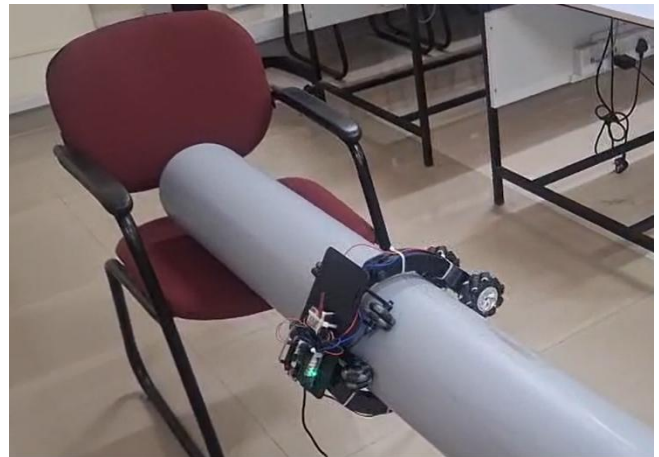
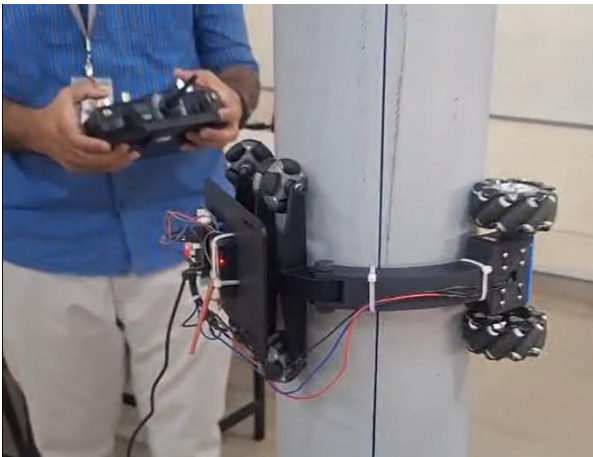


Figure 14 Final prototype during validation testing, mounted on an 8-inch PVC pipe vertically (left) and horizontally (right)

FR1 — Static Grip and Clamping Stability

The robot was mounted onto the vertical test pipe it successfully maintained its position on the pipe under its own static weight without any downward drift, confirming that the hinged arm mechanism generated sufficient normal force at all wheel contact points. The semi-circular frame clamped securely and the Mecanum and Omni wheels maintained continuous and firm contact with the pipe surface throughout the hold duration.

FR2 — Vertical Climbing

Upon issuing vertical climb commands via the RC transmitter, the robot demonstrated controlled upward and downward motion along the pipe axis. The Dynamixel XC430-W240-T motors, operating in velocity control mode, produced sufficient torque to overcome the robot's weight and drive consistently. Both ascending and descending motions were smooth and stable, with all four Mecanum wheels maintaining contact with the pipe surface throughout the run. The transition between upward and downward climbing was responsive, with no observable stalling or slippage, validating the actuator selection based on the motor torque calculation carried out in Section 7.1.3.

FR3 — Circumferential Rotation

The robot successfully demonstrated clockwise and counterclockwise circumferential rotation around the pipe cross-section. Upon issuing rotation commands on the turn channel of the RC transmitter, the differential velocity mixing logic drove the left and right motor pairs in the same direction, generating a tangential force that rotated the robot around the pipe. Rotation was smooth and controlled, with minimal

axial drift observed during the manoeuvre, confirming that the holonomic wheel configuration — specifically the angled roller geometry of the Mecanum wheels — produces the required circumferential force components on the cylindrical surface.

8.3 Key Observations

- **Successful vertical climbing achieved** — the robot climbed vertically and traversed horizontally in a stable and controlled manner, directly validating the effectiveness of the Dynamixel XC430-W240-T actuator selection and the motor torque calculation methodology.
- **Holonomic circumferential traversal demonstrated** — the robot rotated circumferentially around the pipe in both clockwise and counterclockwise directions, confirming the core holonomic capability that differentiates this design from conventional non-holonomic climbing robots.
- **Stable clamping and grip maintained throughout** — the hinged arm mechanism and symmetric wheel arrangement provided consistent contact force at all wheel positions during both vertical climbing and rotational traversal, with no wheel lift-off or loss of grip observed.
- **RC-based wireless control functioned reliably** — the FSR9B receiver and two-channel PWM input system operated without signal dropout during all test runs, with the fail-safe logic correctly defaulting motors to zero velocity upon stick return to neutral.
- **Velocity mixing logic validated** — the differential velocity mixing implemented in the control software correctly translated independent move and turn RC stick inputs into coordinated left and right motor velocity commands, producing the expected robot motions for all four command modes — climb up, climb down, rotate CW, and rotate CCW.
- **Custom motor coupling performed effectively** — the custom 3D printed motor-to-Mecanum-wheel coupling transmitted torque without slippage or backlash, confirming the suitability of the coupling design for the forces involved in vertical pipe climbing.
- **3D printed frame demonstrated structural adequacy** — the PLA frame withstood the mechanical loads during all test conditions without cracking, deformation, or joint failure, validating the structural design.

8.4 Results Summary

Table 4 Summary of functional requirement validation results for Mini Project

Functional Requirement	Test Outcome	Status
FR1 — Static grip without slippage	Robot held position on pipe under self-weight	Pass
FR2 — Vertical climbing and Horizontal Traversal	Controlled ascent and descent achieved	Pass
FR3 — Circumferential rotation (CW and CCW)	Rotation in both directions demonstrated	Pass

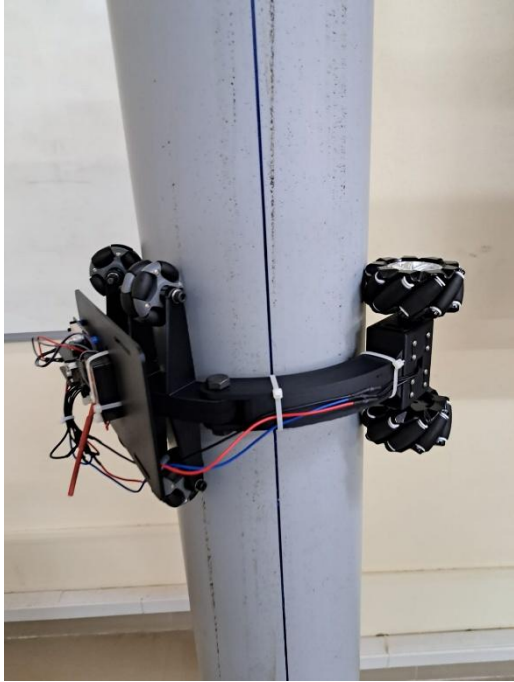


Figure 15 Final prototype mounted on the PVC pipe during vertical climbing validation test

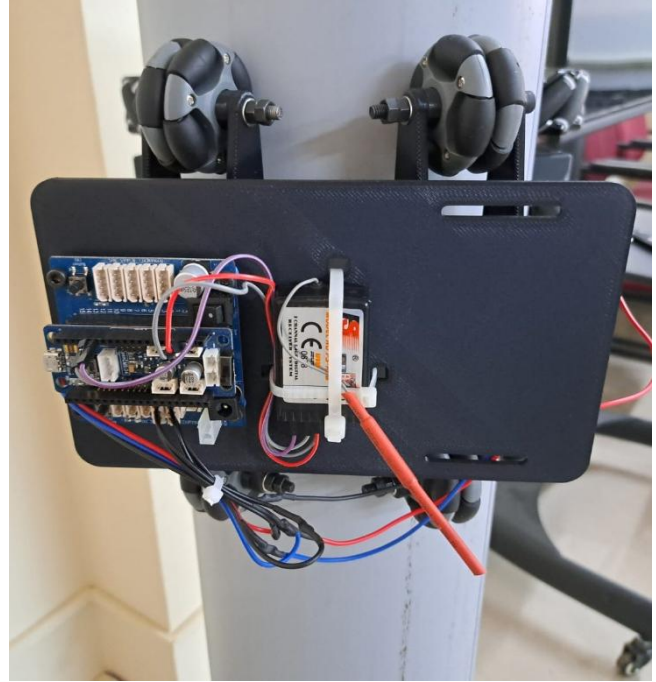


Figure 16 Integrated electronics bay of the final prototype featuring - OpenCM 9.04 microcontroller mounted on OpenCM 485 Expansion Board with FSR9B Receiver alongside

All three functional requirements defined for Mini Project were successfully demonstrated. The final prototype model validated the core design hypothesis that a Mecanum and Omni wheel-based ring-frame robot can achieve holonomic traversal on the outer surface of a cylindrical pipe.

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

This project successfully designed, developed, and validated a Holonomic Pipe Traversal Robot using Mecanum and Omni Wheels — a robotic platform capable of vertical climbing, horizontal traversal and circumferential rotation on the outer surface of a cylindrical pipe, all without mechanical reconfiguration of the robot body.

The development followed a structured, iterative methodology — progressing through conceptual design, analytical motor torque and spring selection calculations, two prototype fabrication and testing cycles, and a final integrated prototype — each stage systematically addressing the limitations of the previous. The transition from TT gear motors in Prototypes 1 and 2 to Dynamixel XC430-W240-T smart servo motors in the final model, guided by the torque calculation, was the decisive factor in achieving reliable vertical climbing.

The final prototype demonstrated all three functional requirements of the Mini Project: stable pipe gripping, controlled vertical climbing in both directions, and circumferential rotation in both clockwise and counterclockwise directions. The RC-based wireless control interface, velocity mixing logic, and closed-loop Dynamixel motor control together provided a responsive, stable, and fail-safe operation framework.

The project establishes a functional proof-of-concept for holonomic external pipe traversal — a capability that is largely absent in existing pipe-climbing robot literature — and lays a concrete technical foundation for its further development into a practical industrial inspection tool.

9.2 Future Scope

The current prototype demonstrates the core traversal/mobility capabilities of the system and opens several meaningful directions for future development:

9.2.1 Sensor Integration for Inspection Applications

The holonomic traversal platform developed in this project provides an ideal base for mounting non-destructive testing and inspection sensors. Future iterations should focus on integrating application-oriented sensor payloads, including:

- **Visual inspection cameras** — for real-time surface imaging, crack detection, and corrosion mapping along the pipe surface
- **Ultrasonic thickness gauges** — for non-contact measurement of pipe wall thickness and detection of internal corrosion or wall thinning
- **Thermal or infrared sensors** — for detecting heat anomalies, insulation defects, or fluid leakage indicators on pipeline surfaces

These sensor integrations would transition the robot from a traversal prototype to a fully functional pipeline inspection system, directly applicable to oil and gas, water supply, and industrial process piping inspection workflows.

9.2.2 Autonomous Navigation and Path Planning

The current system relies on manual RC control for all motion commands. A natural progression would be to implement onboard localisation — using encoder odometry from the Dynamixel motors supplemented by an IMU — and develop autonomous path planning algorithms that enable the robot to systematically cover the entire surface of a pipe section without operator intervention, ensuring complete and repeatable inspection coverage.

9.2.3 Expanded Pipe Diameter Range

The current prototype is optimised for a specific pipe diameter. Future mechanical design iterations should focus on extending the adaptive clamping range through redesigned telescopic arm geometry, enabling the robot to be deployed across the full spectrum of standard industrial pipe sizes — from small-bore process pipes to large-diameter transmission pipelines — without requiring hardware reconfiguration.

9.2.4 Wireless Data Transmission and Remote Monitoring

Integration of a wireless communication module — such as Wi-Fi, Bluetooth or LoRa — would enable real-time transmission of sensor data, motor telemetry, and live camera feeds to a remote monitoring station, supporting deployment in environments that are inaccessible or hazardous to human operators.

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ANNEXURES

Annexure A - Arduino Source Code

```
#include <Dynamixel2Arduino.h>

#define DXL_SERIAL Serial3
#define DEBUG_SERIAL Serial

const int DXL_DIR_PIN = 22;

Dynamixel2Arduino dxl(DXL_SERIAL, DXL_DIR_PIN);

const float DXL_PROTOCOL_VERSION = 2.0;

// RC Receiver pins
#define MOVE_CH 10 // CH2 -> Up / Down
#define TURN_CH 11 // CH4 -> Rotate CW / CCW

int movePWM = 1500;
int turnPWM = 1500;

void setup() {

  DEBUG_SERIAL.begin(115200);

  pinMode(MOVE_CH, INPUT);
  pinMode(TURN_CH, INPUT);

  // Dynamixel setup
  dxl.begin(1000000);
  dxl.setPortProtocolVersion(DXL_PROTOCOL_VERSION);

  // Configure motors
  for (int DXL_ID = 1; DXL_ID <= 4; DXL_ID++) {
    dxl.torqueOff(DXL_ID);
    dxl.setOperatingMode(DXL_ID, OP_VELOCITY);
    dxl.torqueOn(DXL_ID);
  }
  DEBUG_SERIAL.println("Dual Channel Dynamixel Control Started");
}

void loop() {
  // Read PWM
  movePWM = pulseIn(MOVE_CH, HIGH, 25000);
  turnPWM = pulseIn(TURN_CH, HIGH, 25000);

  // Fail-safe
  if (movePWM < 1000 || movePWM > 2000)
    movePWM = 1500;
  if (turnPWM < 1000 || turnPWM > 2000)
    turnPWM = 1500;
```

```
// Convert PWM to RPM
float moveVelocity = pwmToRPM(movePWM);
float turnVelocity = pwmToRPM(turnPWM);

// Mixing
float leftVelocity = -moveVelocity + turnVelocity;
float rightVelocity = moveVelocity + turnVelocity;

// Limit RPM
leftVelocity = constrain(leftVelocity, -40, 40);
rightVelocity = constrain(rightVelocity, -40, 40);

// Debug
// Debug
DEBUG_SERIAL.print("MOVE PWM: ");
DEBUG_SERIAL.print(movePWM);

DEBUG_SERIAL.print(" | TURN PWM: ");
DEBUG_SERIAL.print(turnPWM);

DEBUG_SERIAL.print(" | MOVE RPM: ");
DEBUG_SERIAL.print(moveVelocity);

DEBUG_SERIAL.print(" | TURN RPM: ");
DEBUG_SERIAL.print(turnVelocity);

DEBUG_SERIAL.print(" | LEFT RPM: ");
DEBUG_SERIAL.print(leftVelocity);

DEBUG_SERIAL.print(" | RIGHT RPM: ");
DEBUG_SERIAL.println(rightVelocity);

// Left side motors
dxl.setGoalVelocity(1, leftVelocity, UNIT_RPM);
dxl.setGoalVelocity(2, leftVelocity, UNIT_RPM);

// Right side motors
dxl.setGoalVelocity(3, rightVelocity, UNIT_RPM);
dxl.setGoalVelocity(4, rightVelocity, UNIT_RPM);

delay(20);
}

// =====
// Function: PWM -> RPM with deadband
// =====

float pwmToRPM(int pwm)
{
    // Deadband
    if (pwm >= 1460 && pwm <= 1540) {
        return 0;
    }
}
```

```
// Reverse
else if (pwm < 1460) {
    return map(pwm, 1000, 1460, -40, 0);
}

// Forward
else {
    return map(pwm, 1540, 2000, 0, 40);
}
}
```